

Exercise 1: Control Structures

Aim

To implement banking operations using MySQL queries, including updating customer records and retrieving loan information based on given conditions.

```
CREATE DATABASE BankDB;
```

```
USE BankDB;
```

```
CREATE TABLE Customer(  
    CustomerID INT PRIMARY KEY,  
    CustomerName varchar(50),  
    Age INT,  
    Balance decimal(10,2),  
    LoanInterestRate decimal(5,2),  
    IsVIP boolean  
);
```

```
CREATE TABLE Loans(  
    LoanID INT PRIMARY KEY,  
    CustomerID INT,  
    LoanAmount DECIMAL(10,2),  
    LoanDueDate DATE,  
    FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID)  
);
```

Insert into Customer Values

```
(101,'Krishna',65,15000,9.5,FALSE),
(102,'Rahul',45,8000,10.0,FALSE),
(103,'Priya',70,20000,8.5,FALSE),
(104,'Arun',35,12000,11.0,FALSE),
(105,'Sneha',62,5000,9.0,FALSE);
```

```
INSERT INTO Loans VALUES
```

```
(1,101,500000,'2026-07-15'),
(2,102,300000,'2026-09-10'),
(3,103,400000,'2026-07-20'),
(4,104,250000,'2026-08-25'),
(5,105,350000,'2026-07-10');
```

```
SELECT * FROM Customer;
```

	CustomerID	CustomerName	Age	Balance	LoanInterestRate	IsVIP
▶	101	Krishna	65	15000.00	3.50	1
	102	Rahul	45	8000.00	10.00	0
	103	Priya	70	20000.00	2.50	1
	104	Arun	35	12000.00	11.00	1
	105	Sneha	62	5000.00	3.00	0

--**Question:** Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

```
UPDATE Customer SET LoanInterestRate = LoanInterestRate - 1 WHERE Age > 60;
```

```
SELECT * FROM Customer;
```

	CustomerID	CustomerName	Age	Balance	LoanInterestRate	IsVIP
▶	101	Krishna	65	15000.00	2.50	1
	102	Rahul	45	8000.00	10.00	0
	103	Priya	70	20000.00	1.50	1
	104	Arun	35	12000.00	11.00	1
	105	Sneha	62	5000.00	2.00	0

--**Question:** Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over \$7,000.

```
UPDATE Customer SET IsVIP = TRUE WHERE Balance > 7000;
```

```
SELECT * FROM Customer;
```

	CustomerID	CustomerName	Age	Balance	LoanInterestRate	IsVIP
►	101	Krishna	65	15000.00	2.50	1
	102	Rahul	45	8000.00	10.00	1
	103	Priya	70	20000.00	1.50	1
	104	Arun	35	12000.00	11.00	1
	105	Sneha	62	5000.00	2.00	0

--**Question:** Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.

```
SELECT
```

```
    C.CustomerName,
```

```
    L.LoanDueDate
```

```
FROM Customer C
```

```
JOIN Loans L
```

```
ON C.CustomerID = L.CustomerID
```

```
WHERE L.LoanDueDate
```

```
BETWEEN '2026-07-01' AND '2026-07-31';
```

	CustomerName	LoanDueDate
►	Krishna	2026-07-15
	Priya	2026-07-20
	Sneha	2026-07-10